

Ashwanth Senthil Kumar

Software Engineer | AI/ML Developer

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Computer Science Engineering student (AI/ML Honors) with demonstrated expertise in full-stack development and applied machine learning. Seeking a software engineering internship or entry-level SDE/ML role where proven ability to deliver production-quality, AI-driven solutions.

INTERNSHIP EXPERIENCE

Qono Technologies | Machine Learning Intern

May 2025 – June 2025

Onsite | Coimbatore | India

- Engineered end-to-end machine learning systems, developing a production ready IoT malware classifier using Random Forest and SVM alongside scalable recommendation engines powered by collaborative filtering and TF-IDF.

Issential Solutions | AI/ML Intern

December 2025

Onsite | Coimbatore | India

- Integrated Explainable AI (SHAP, LIME) into classification pipelines to optimize model interpretability, engineering visual feature-attribution reports (waterfall and force plots) that translated complex black-box predictions for key stakeholders.

EDUCATION

Sri Ramakrishnan Engineering College (SREC)

Bachelor of Engineering - Computer Science | CGPA: 7.97/10

Honors specialization in Artificial Intelligence & Machine Learning

Coimbatore, Tamil Nadu

August 2023 – Present

June 2025 - Present

Bharani Vidhyalaya Sr. Sec. School (CBSE)

HSC | Percentage: 80%

SLSC | Percentage: 75%

Karur, Tamil Nadu

June 2022 – May 2023

June 2020 – May 2021

SKILLS

Languages: Python, Java, JavaScript, R

Frontend: ReactJS, HTML5, CSS3

Backend & APIs: FastAPI, REST API Design, API Integration, JSON

AI/ML: TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy, LLM Integration (OpenAI, Gemini), Hugging Face, Prompt Engineering, Feature Engineering, Model Evaluation

Databases: MySQL, PostgreSQL, MongoDB

DevOps & Tools: Git, GitHub, GitHub Action, Docker, Tableau

Core Concepts: Data Structures & Algorithms, OOPs, DBMS, Cloud Computing (AWS, GCP Vertex AI), System Design, Agile/Scrum Development, MLOps

PROJECTS

Explainable AI for Customer Churn Prediction and Retention Strategy | XGBoost • XAI • MLOps | [GitHub](#)

- Engineered an end-to-end churn prediction pipeline using **XGBoost** on customer behavioral and transactional data, addressing imbalance data leakage to achieve robust precision-recall performance on a highly skewed dataset (~5% churn rate).
- Integrated **XAI (Explainable AI)** to deliver interpretable, explanations for individual churn predictions, translating outputs into actionable **retention strategies** and enabling business interventions over generic blanket offers.

Recipe Archaeologist - AI-Powered Visual Ingredient Reconstruction | OpenCV • LLM • ReactJS | [GitHub](#)

- Architected a multimodal pipeline integrating computer vision (OpenCV) and prompt-engineered **AI Agents and LLMs** to accurately derive ingredient probabilities from complex visual patterns.
- Executed **Full Stack Development** by building a robust FastAPI backend and a ReactJS frontend optimized for seamless **UI / UX Design**, delivering end-to-end image-to-recipe outputs in under 10 seconds.

CERTIFICATION

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|---|--------------------|
| • Machine Learning Specialization – Coursera Stanford University (Andrew Ng) | June 2025 |
| • Deep Learning – NPTEL (IIT Ropar) | April 2025 |
| • Introduction to Large Language Model – NPTEL (IIT Madras) | April 2026 |
| • Citizen Data Science Certification — Infosys Springboard | August 2025 |

PUBLICATION & ACTIVITIES

- **First Author Publication (Springer):** *Supply Chain Forecasting for Inventory Management* – International Conference on Data Analytics and Insights (ICDAI 2025).
- **Leadership:** Chairperson, IEEE Computational Intelligence Society (SREC Student Branch Chapter).
- **Workshop & Induction Program Lead, IEEE:** Planned, conducted, and managed technical workshops and induction programs for IEEE members.
- **IEEE Xtreme 19.0 Hackathon:** Contributed to design coordination and event management for a 24-hour global hackathon.